

Requirements Table — Academic Elective Bidding & Allocation System

Campus & Academic Operations | Actors: Student, Academic Registrar, Timetable System

PES1UG24CS610

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites validated; Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Ensures bids reflect genuine student preference strength and that only eligible students compete for a seat.	
FR-002	Functional	The system shall allow a student to rank at least 3 and at most 8 elective preferences in strict order before submitting a bid.	High	Pass: Submission is accepted only when between 3 and 8 uniquely ranked electives are selected; Fail: Duplicate ranks or fewer than 3 electives are rejected with an inline error.	A minimum and maximum preference count keeps the solver's search space bounded and prevents trivial single-choice bidding.	
FR-003	Functional	The system shall run an allocation solver that assigns electives to students based on bid amount, prerequisite eligibility, and seat capacity, while eliminating any resulting timetable clashes.	High	Pass: Every published allocation has zero students assigned to two electives sharing the same time slot and no elective exceeds its seat cap; Fail: Any capacity or clash violation exists.	The core value of the system is a conflict-free, capacity-respecting allocation, not merely tallying bids.	
FR-004	Functional	The system shall allow the Academic Registrar to review the solver's proposed allocation and manually override at most one assignment per student before publishing.	Medium	Pass: Registrar can view proposed allocations and successfully apply an override that is reflected in the final list; Fail: Override is silently dropped or applied without an audit record.	Automated solvers cannot capture every edge case (e.g., medical exemptions), so a bounded human override is required.	
FR-005	Functional	The system shall notify each student of their final elective allocation via portal notification and email within 5 minutes of the Registrar publishing results.	Medium	Pass: 100% of students receive a notification within 5 minutes of publish, logged with a timestamp; Fail: Any student is missed or notified after the 5-minute window.	Timely notification lets students plan their semester and raise grievances within the appeal window.	
NFR-001	Nonfunctional (Performance & Security)	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	The bidding window is short and university-wide, so the solver must return results within a single peak session.	
NFR-002	Nonfunctional (Security)	The system shall restrict bid submission and viewing of allocation results to the authenticated owning student, using the university SSO, with all bid and override actions logged for audit.	High	Pass: Penetration test confirms a student cannot view or alter another student's bid, and every bid/override action appears in the audit log with actor ID and timestamp; Fail: Any unauthorized access or missing log entry.	Bidding credits and allocation outcomes affect academic standing, so tampering or impersonation must be prevented and traceable.	